

Project Proposal: Raster to Vector Image Vectorizer with Curve Fitting

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1. Project Goal

User Perspective: I will build a tool that allows a user to input a standard raster image (such as a PNG) and automatically converts it into a scalable vector graphic (SVG) format.

Main Objective: The primary goal is to trace the raw pixel outlines of a given image, systematically reduce the number of vertices within the resulting paths, and apply smooth Bézier curves to the simplified geometry.

2. Core Geometric Concepts

Algorithms: The core of this project relies on implementing polyline simplification algorithms, specifically the Ramer-Douglas-Peucker algorithm, to reduce the complexity of the initial traced outlines. I will also implement the de Casteljau algorithm to process and render smooth Bézier curves over the simplified shapes.

Why these are the right tools: The Ramer-Douglas-Peucker algorithm is essential for removing redundant vertices while preserving the fundamental topology of the traced shapes. The de Casteljau algorithm provides a robust geometric method for creating aesthetically pleasing, smooth curves from the remaining simplified control points.

3. Scope & Key Features (The “Finish Line”)

Minimum Viable Product (MVP):

- Load a raster image file (e.g., PNG) into the program.
- Extract raw geometric paths by tracing pixel outlines.
- Successfully simplify the generated polylines to reduce vertex count.

Stretch Goals (Optional):

- Implement the de Casteljau algorithm to replace rigid polylines with smooth Bézier curves.
- Add functionality to export the final calculated geometry as a valid SVG file.